

Mahsa Movagh Behrouzi

Location: Tehran, Iran

GitHub | Aparat | Email: mahsamovagh@gmail.com | Mobile: +98 914 242 6370

PROFESSIONAL SUMMARY

Data Scientist with **1.5 years of industry experience** at **Baam (Sadad Informatics Corporation)** and a Master's degree in Computer Science from Iran University of Science and Technology (IUST). Proficient in **PyTorch**, **NumPy**, and **Pandas**, with end-to-end experience across the ML lifecycle: data collection, preprocessing, feature engineering, model training, and evaluation. Hands-on industry experience with **Automatic Speech Recognition (ASR)**, **Large Language Models (LLMs)**, and **Retrieval-Augmented Generation (RAG)**. Academic experience in **Reinforcement Learning**, including **DQN** and **PPO**, with a strong interest in developing practical AI and machine learning solutions.

EXPERIENCE

Baam Digital Banking Platform

Data Scientist

Feb. 2025 – Present

- Fine-tuned a **large-scale speech recognition model** for Persian Automatic Speech Recognition, achieving **17.38% WER** on a real-world Persian dataset compared with **29.4% and 37.4% WER** for baseline models on widely used benchmark datasets, while developing end-to-end preprocessing, training, and evaluation pipelines.
- Developed a **Persian-to-English transliteration model** for customer name normalization, which was deployed in production to support an English billing-statement feature.
- Conducted **anomaly detection and behavioral analysis** on banking platform data to identify unusual user activity and abuse patterns; identified a root cause responsible for **78% of password-creation failures**, leading to a product fix that reduced failed attempts by **80%**.
- Deployed open-source **Large Language Models (LLMs)** locally for automated text post-processing and **synthetic dataset generation** across multiple machine learning projects.
- Worked across the full **machine learning lifecycle**, including **data collection, preprocessing, feature engineering, model training, evaluation, and deployment**, while collaborating with cross-functional teams to deliver production ML solutions.

TECHNICAL SKILLS

Programming and Framework

Python, Flask

Machine Learning

PyTorch, Scikit-Learn, NumPy, Pandas

AI

Whisper, Vosk, Gemma, Ollama, RL (DQN, PPO), RAG, Agentic AI, LLM-based Applications

Databases

Elasticsearch, MySQL

Tools

Docker, Git, Linux, VS Code

Platforms

Hugging Face, Kaggle

EDUCATION

Iran University of Science and Technology

Master of Science in Computer Science

Sep 2022 – Oct 2025

University of Zanjan

Bachelor of Science in Computer Science

Sep 2015 – Sep 2020

ACADEMIC PROJECTS

Pattern Recognition Course

Supervisor: Dr. Morteza Analoui

- Implemented **Crammer-Singer SVM** from scratch and applied it for classification on the MNIST dataset.
- Utilized **Adaptive Boosting (AdaBoost)** and **XGBoost** for boosting techniques.
- Developed a **CNN-based classifier** on the Persian Traffic Sign Dataset (PTSD).
- Explored meta-learning techniques with **Model-Agnostic Meta-Learning (MAML)** and **MetaOptNet**.

Deep Learning Course

Supervisor: Dr. MohammadReza Mohammadi

- Implemented **linear regression** and **logistic regression** models from scratch.
- Developed **MLP**, **CNNs** for classification, and visualized kernels and feature maps.
- Created an **image captioning** model using LSTMs.
- Implemented a **Vision Transformer** on the Linnaeus 5 dataset.
- Gained familiarity with **Optuna** for hyperparameter tuning.
- Explored **GANs**, **Conditional GANs**, **Cycle GANs**, and **causal learning**.

Software Architectures Course

Supervisor: Dr. Mehrdad Ashtiani

- Collaborated with a team to implement methodologies from the research paper [Product line architecture recovery with outlier filtering in software families: the Apo-Games case study](#).
- Developed core functionality to extract package-level data and construct the **Design Structure Matrix (DSM)** to represent the development view of the architecture.

Bachelor's Final Project

Credit Card Fraud Detection

- Designed and implemented a **Credit Card Fraud Detection** system using **Convolutional Neural Networks (CNNs)**.
- Worked on data preprocessing, feature representation, and model evaluation for highly imbalanced transaction data.
- This project marked my first hands-on experience with **neural networks**.

Master's Final Project

LLM and Reinforcement Learning Integration

- Proposed a hybrid framework combining **Large Language Models (LLaMA3-8B, GPT-4o mini)** with **Reinforcement Learning (DreamerV3)** to enhance decision-making agents.
- Designed and evaluated **serial, parallel, and hybrid integration strategies** in the **Craier** game environment to leverage both textual and experiential knowledge.

TEACHING ASSISTANCE

Artificial Intelligence

Instructor: Dr. MohammadReza Mohammadi

Sep 2023 – Jan 2024

Fundamentals of Deep Learning

Instructor: Dr. Marzieh Davoodabadi

Sep 2023 – Jan 2024

Deep Learning

Instructor: Dr. MohammadReza Mohammadi

Feb 2024 – Jun 2024

Pattern Recognition

Instructor: Morteza Analoui

Sep 2024 – Jan 2025